

Improving Agricultural Yield Predictions: An Examination of Existing Methods and Development of Novel Models

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Management Summary

As part of our Financial Engineering senior project, we collaborated with Watts and Associates to enhance crop yield forecasting for crop insurance. Our objective was to optimize the accuracy and efficiency of crop yield predictions, ensuring better risk assessment and management for agricultural stakeholders. By implementing advanced modeling techniques and leveraging historical data, we aimed to provide more reliable forecasts for crop insurance providers and farmers alike.

Accurate county yield predictions have far-reaching implications for various stakeholders, including crop producers, insurance companies, and consumers. Underestimated yields may lead to inadequate coverage for farmers, while overestimations can result in excessive indemnity payouts. By developing a more precise forecasting model, this project aims to provide valuable insights and recommendations to W&A, enabling them to better advise their clients and contribute to the overall well-being of the agricultural community.

The project was carried out in three phases. Phase 1 involved replicating the RMA's forecasting model, which uses a linear piecewise double-knot spline method, and evaluating its forecast accuracy through out-of-sample testing over individual counties within three states. Phase 2 focused on replicating Watts and Associates' forecasting model, which operates on the same underlying model as phase 1, but uses defined risk regions rather than individual counties. Phase 3 was dedicated to creating an innovative model that leverages the team's expertise in statistics and forecasting to improve upon the existing models.

In innovating our new forecasting techniques, our first adjustment involved strategically altering the placement constraints of splines within regression models, particularly focusing on the final spline. Traditionally, this last spline was positioned within a narrow window near the dataset's end, usually extending beyond the last five years. However, recognizing the need for a nuanced approach to agricultural dynamics, the team reduced the permissible range for spline placement. Specifically, we modified the constraint so that the final spline must be positioned further from the dataset's end, with a minimum distance of 10 and 15 years. This adjustment was motivated to combat overfitting and enhance model generalizability. By anchoring the final spline to data points further from the present, the models prioritize capturing long-term trends while minimizing the impact of short-term fluctuations or noise, thereby improving the accuracy of crop yield predictions. In some situations this implementation reduced the forecast error by close to 40%.

Another significant enhancement implemented by the team involves revising the fundamental structure of the regression models. Acknowledging the complex interplay of technological advancements, agronomic practices, and environmental influences in agricultural production, the team incorporated a crucial modification stipulating that the final slopes within the regression framework must be non-negative. This adjustment reflects an understanding of agricultural dynamics, recognizing the persistent upward trajectory of crop yields driven by historical advancements in agricultural science and technology. This strategic modification not only ensures model accuracy but also facilitates a clearer understanding of how technological innovations and agricultural interventions contribute to yield enhancements. When using this model the forecast error was reduced by close to 40% as well.

In conclusion, our two variations on the current yield forecasting method could result in higher predictive capabilities. If this increased accuracy holds across other agricultural areas, implementing the two models could result in increased benefits for producers, insurers, and consumers.

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1. Problem Overview

Background

Watts and Associates (W&A), situated in Billings, Montana, is a consultant within the agricultural insurance sector. Established by Tim Watts in the 1980s, W&A was established with the mission of enhancing the well-being of farmers and ranchers by delivering valuable benefits. The consultant agency operates collaboratively with the following key stakeholders: the Risk Management Agency (RMA), producer groups, agencies, as well as insurance and reinsurance companies. The Risk Management Agency, a pivotal division of the United States Department of Agriculture, spearheads the nation's crop insurance industry. It plays a crucial role not only in forecasting expected yields for various crops across their respective growing locations but also in overseeing the Federal Crop Insurance Corporation program, thereby ensuring that crop insurance remains both accessible and affordable to agricultural producers. These forecasts are essential for determining the premiums of subsidized crop revenue insurance plans. In 2023, the RMA was responsible for projecting yields for over 15,000 crop-county combinations, facilitating the calculation of indemnities on more than \$16 billion in insurance premiums. The accuracy of these county-specific yield forecasts is critical, impacting both the producers, who rely on these policies for financial protection, and the insurance companies that provide them. Discrepancies in forecasting can lead to significant financial consequences: underpredicted yields may result in inadequate coverage for farmers, while overestimations can trigger extensive indemnity payouts. To mitigate such risks, the RMA provides reinsurance to insurers, ensuring that losses exceeding certain thresholds are managed effectively. This safeguard, however, implicates taxpayer funds, highlighting the need for precise and equitable forecasting practices. The balance RMA seeks to maintain between providing adequate insurance coverage and ensuring fiscal responsibility underscores the agency's vital role in the agricultural sector's resilience. The challenge this project aims to tackle is to provide an in-depth analysis and evaluation of the county yield predictions to provide Watts and Associates with insights into the overall accuracy and reliability of the Watts and Associates' and the RMA's forecast to best advise their clients. The success of this project will not only benefit producers and insurance companies but will also contribute valuable insights to the broader agricultural community.

In the realm of agricultural insurance, the accuracy of crop yield predictions stands as a linchpin for ensuring equitable compensation for farmers and prudent allocation of public funds. The main trigger for insurance payments hinges upon the reliability of forecasted crop yields. These predictions serve as a crucial tool for insurance providers to assess the extent of losses incurred by farmers due to factors such as adverse weather conditions, pest infestations, or other unforeseen circumstances. The significance of accurate crop yield forecasts cannot be overstated. For farmers, timely and precise insurance payouts can mean the difference between financial stability and economic hardship in the face of crop failure. Conversely, for government entities, such as the United States Department of Agriculture, responsible for overseeing agricultural insurance programs, ensuring the accuracy of yield predictions is imperative to prevent overpayment and safeguard taxpayers' money. By relying on robust forecasting models, insurance providers can effectively tailor their payouts to match the actual losses experienced by farmers, thereby minimizing both under-compensation and excessive payouts. Moreover, accurate crop yield predictions contribute to the overall sustainability and viability of agricultural insurance programs. By instilling confidence among farmers in the reliability of insurance coverage, precise forecasts encourage greater participation in these risk management schemes. This, in turn, fosters resilience within the agricultural sector by enabling farmers to make informed decisions regarding investments, input usage, and crop planning. In essence, the relevance of accurate crop yield predictions to the insurance industry extends beyond mere financial considerations. It underscores the broader objectives

of ensuring food security, mitigating rural poverty and promoting sustainable agricultural practices. As such, continued research and investment in improving the precision of yield forecasting models are imperative to meet the evolving needs of farmers and policymakers in an increasingly dynamic agricultural landscape. The graph below illustrates the relationship among indemnity payouts, total insurance premiums received by insurance companies, and the premiums paid by agricultural producers.

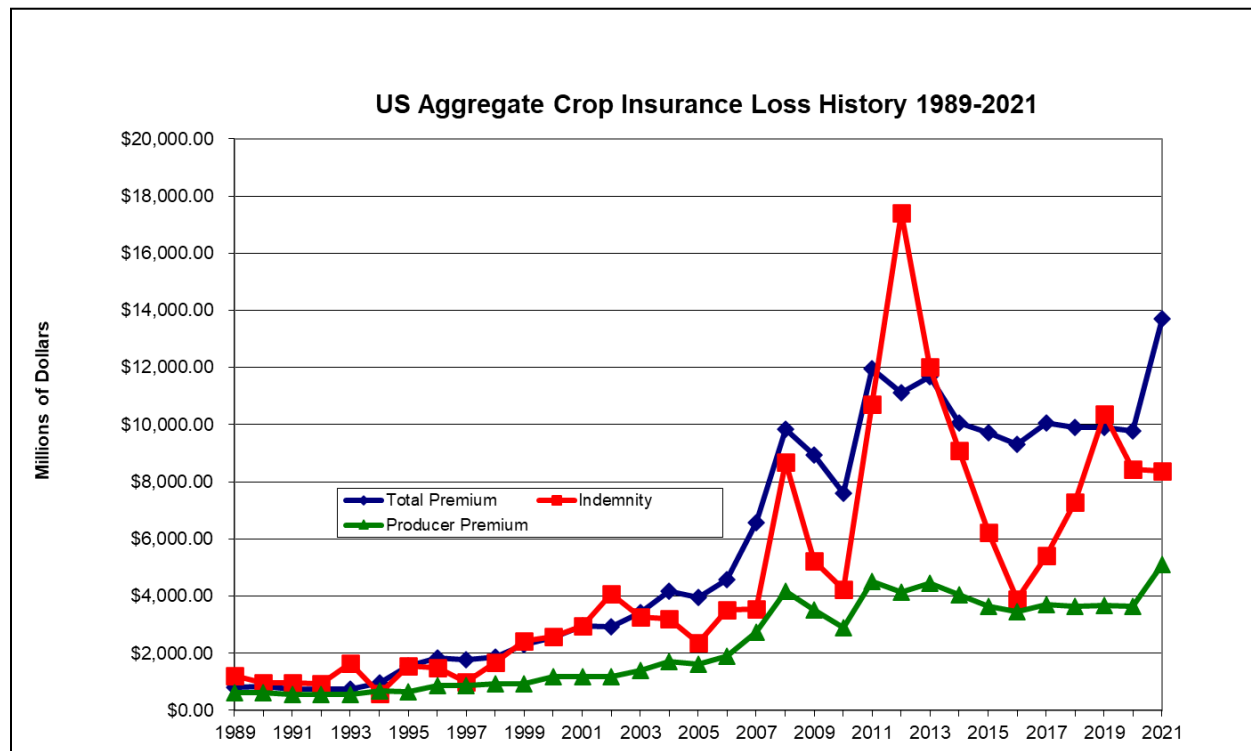


Figure 1. Time Series Showing the Relationship Between Premium and Indemnity.

Objectives

The project encompasses three primary objectives: firstly, to evaluate the historical accuracy of county yield predictions made by the Risk Management Agency; secondly, to assess the predictive capabilities of the model developed by Watts & Associates for county yields; and thirdly, to endeavor in creating a novel model aimed at surpassing predictions made by both the RMA and W&A. Replication of the RMA and W&A models is integral to gaining a comprehensive understanding of the underlying processes and data interpretation. Utilizing the data provided by W&A, which includes corn and soy yields from selected counties, our team will replicate these models. The evaluation of model accuracy will involve the computation of out-of-sample root mean squared forecast error (RMSFE).

The collaborative effort of our team, comprising Merlin McKean, Hans Petersen, Tommy Mellott, and Max Taylor, will be crucial in the creation of our predictive model. Employing the tools at our disposal, the team will strive to minimize the out-of-sample root mean squared forecast error, ensuring the development of a model with optimal accuracy.

Scope

The project's scope revolves around the development of a predictive model using the R programming language to accurately forecast county crop yields for Watts & Associates. Central to this endeavor is the utilization of data-driven techniques aimed at enhancing the precision and reliability of yield predictions. By focusing on county-level data and employing statistical methodologies, the project seeks to facilitate informed decision-making in agricultural risk management. Through comprehensive analysis and modeling, the aim is to provide stakeholders with valuable insights into crop yield dynamics, thereby enabling proactive measures to mitigate risks and optimize agricultural outcomes.

Assumptions

In undertaking this project, several key assumptions guide the modeling process effectively. Firstly, it is assumed that the project will have access to comprehensive and reliable datasets containing historical county-level crop yield data, weather patterns, soil characteristics, and other relevant factors necessary for model training and validation. Secondly, the suitability of the chosen statistical methodologies, such as Ordinary Least Squares (OLS) regression and Piecewise Linear Regression, implemented in R, is assumed to adequately capture the complex relationships between various predictor variables and crop yields, thus yielding accurate predictions. These assumptions are founded on the premise that OLS regression and Piecewise Linear Regression are appropriate for modeling linear and piecewise linear relationships, respectively, within the dataset.

Limitations

While the project aims to develop a highly accurate predictive model, several inherent limitations need to be acknowledged. Firstly, the quality and completeness of available data, including inaccuracies, missing values, and inconsistencies, may impact the reliability and generalizability of the predictive model, regardless of the statistical techniques employed. Additionally, while OLS regression and Piecewise Linear Regression are valuable tools for modeling linear and piecewise linear relationships, respectively, within the dataset, they may not capture nonlinear relationships or interactions among variables, potentially limiting the model's predictive capabilities. Furthermore, the complexity of the model may introduce challenges related to interpretability, computational resources, and overfitting, particularly when using advanced regression techniques. Additionally, external factors such as market dynamics, policy changes, and unforeseen events (e.g., natural disasters) are beyond the scope of the predictive model but may significantly impact crop yields and the accuracy of predictions.

Moreover, the project's geographical scope is limited to counties within the states of South Dakota, Kansas, and Iowa. While these states offer valuable insights into agricultural dynamics, the exclusion of counties from other regions may limit the generalizability of the predictive model to a broader geographical context. Variability in agricultural practices, soil conditions, climate patterns, and other geographical factors across different states may influence crop yields differently, thus affecting the model's applicability beyond the specified region. This limitation underscores the need for cautious interpretation and potential adjustments when extrapolating the model's findings to other geographic areas.

2. Methods

Phase 1

In Phase 1 of the study, the process began with establishing familiarity with the crop insurance system by replicating the model used by the RMA. The RMA's model focuses on forecasting the yield of the crop of interest for each county. The methodology followed a structured outline. Each county was segmented into 30-year periods, such as from 1947 to 1976, from 1948 to 1977, etc. Within each period, a series of steps were executed, each involving either zero, one, or two splines. A spline can be visualized as a pivot point of a piecewise function. A piecewise line with three segments will have two splines, a line with two segments has one spline, and a zero-spline line is simply a straight line. Initially, Ordinary Least Squares (OLS) regression was performed on the time series data. Because OLS seeks to fit a line of best fit, this was known as the zero-spline model.

Subsequently, a piecewise linear single spline forecast was attempted on the same time series, which placed one pivot point to produce lines with two segments. According to W&A's stipulations, constraints were implemented prohibiting splines from occurring within the first or last five years of each 30-year window. For each valid breakpoint, an indicator variable was added to the model to let the slope change after the breakpoint. The sum of squared errors (SSE) of every model over the defined timespan was collected, with the optimal year for each breakpoint being that with the lowest SSE.

Through the use of a similar process, a piecewise linear double-spline model was constructed. This model was also under the constraint of limiting breakpoints to outside the first or last 5 years. Using the same method of iterating through dummy variables, each combination of breakpoint years was used to build a set of models. Again, the model with the lowest SSE value was chosen as the optimal model.

Once the optimal zero, single, and double spline models were obtained for each window, the Akaike Information Criterion (AIC) values were computed for each. The AIC is a goodness-of-fit statistic that allows for a fair comparison of models with a different amount of variables by adding a penalty term for the complexity of the model. A model with many terms, even if it can capture large amounts of variation and correlation between factors, may not outperform a simpler model based on this criterion. The model with the lowest AIC value - the best balance between accuracy and complexity - was used to calculate the final forecast of a given crop-county combination.

The fundamental goal of such forecasting is to predict the final value of the yield variable two years past the end of the 30-year window. As such, the optimal model using the processes and criteria above was used to extrapolate each yield from the end of the 30-year forecast window to two years in the future.

To assess the forecasting accuracy of this extrapolation, out-of-sample tests were performed through the use of backtesting, measuring the disparity between the forecasted yield and the observed yield for the corresponding year for each of the optimized models. Using these errors, the Root Mean Squared Forecast Error (RMSFE) was calculated as a metric of forecasting accuracy for each crop/county/year model combination. The graph below gives a visual of the three regression techniques for Harden, Iowa for corn in the window from 1947-1976. The black dotted line indicates when in-sample data transitions to historical backtesting.

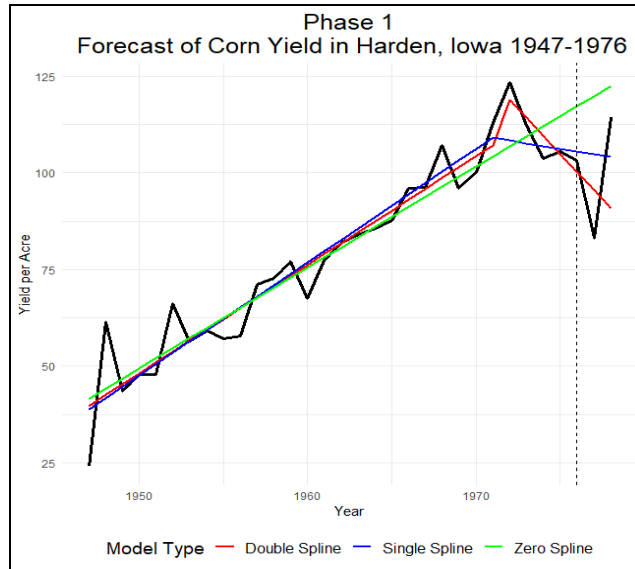


Figure 2. Visual of Regression Techniques Used in Harden, Iowa from 1947-1976.

Phase 2

In Phase 2, the primary objective was to enhance comprehension of and replicate the existing methodologies utilized by Watts and Associates. The approach employed by W&A involved grouping counties within a state into Risk Regions based on historical similarities among them. A visual depiction of the risk regions within Illinois is shown in the figure below.

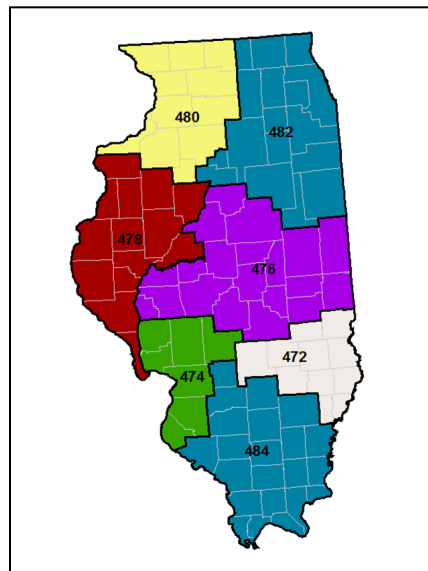


Figure 3. Risk Region Division in Illinois

The yield per acre of the aggregated risk region was determined by the weighted average yield per acre of all counties within the risk region. Subsequently, forecasting at Watts and Associates was conducted

based on the aggregated risk region data. The forecast was then adjusted to fit the individual counties within the risk region by vertically shifting it, taking into account the mean difference between the observed yield per acre in the entire risk region and the yield per acre in each county. The underlying assumption of Watts and Associates is that when similar counties are aggregated together, the underlying trend is more heavily focused. A visual of how these aggregated counties may eliminate the effect of random occurrences is shown in the figure below.

To forecast the aggregated risk region, the weighted average yield per acre in the risk region was calculated. For each risk region, the data was subsetted into a 30-year window. Ordinary Least Squares (OLS) regression analysis was performed on the time series data using a zero spline. A piecewise linear single spline forecast was then applied to the time series data, with the same constraints as described above. The forecast was rerun at each permissible year, and the in-sample Sum of Squared Errors (SSE) was measured. The model with the lowest in-sample SSE, known as the optimized single spline model, was retained.

The methodology was repeated for the piecewise linear double spline forecast, with both splines constrained from occurring within the initial or final 5 years of the 30-year window. The forecast was rerun with all possible combinations of splines within the allowable window, and the model with the lowest in-sample SSE, the optimized double spline model, was selected. Similarly to Phase 1, the Akaike Information Criterion (AIC) values were calculated and used to find the best model for the crop produced in a risk region, over the given timeframe. The optimal model was used to calculate the forecasted yield per acre of a crop for the entire risk region.

To transform the risk region model back to the county level, the mean difference between the aggregated risk region's weighted average yield per acre and the actual yield per acre in each county was determined over the 30-year window. The forecasted value for each county was then vertically shifted based on this mean difference. A visual depiction of the final vertical shift is shown in the figure below.

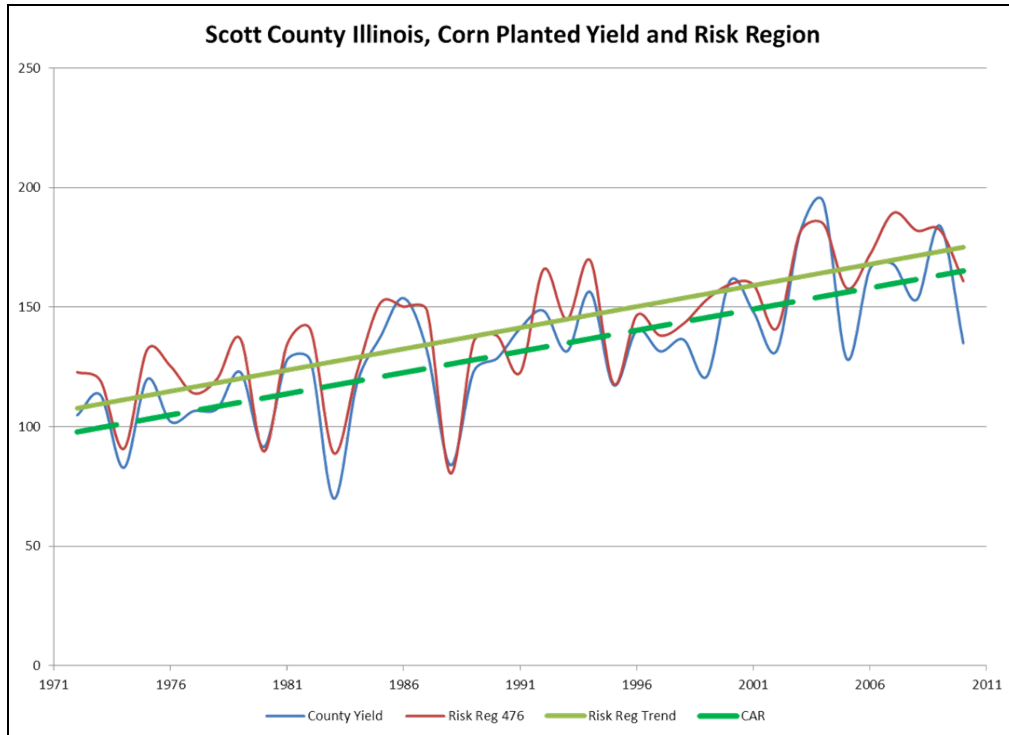


Figure 4. Visual Depiction of Vertically Shifting from Risk Region Forecast to County Forecast

This adjusted forecasted value for each county was utilized in the same backtesting procedure to calculate the error between the model's forecasted yield and the actual observed yield, on the county level. This iterative process was conducted for each county and crop combination within a 30-year window, with the window advancing by one year in each iteration, until all combinations were exhaustively assessed. The entire procedure was repeated for each available risk region, ensuring a comprehensive evaluation. Finally, the errors were used to calculate the Root Mean Squared Forecast Error (RMSFE) as a metric to gauge the accuracy of the forecast for each crop/county/year model combination. The graph below gives a visual of the three regression techniques for Harden, Iowa for corn in the window from 1947-1976. The black dotted line indicates when in-sample data transitions to historical backtesting.

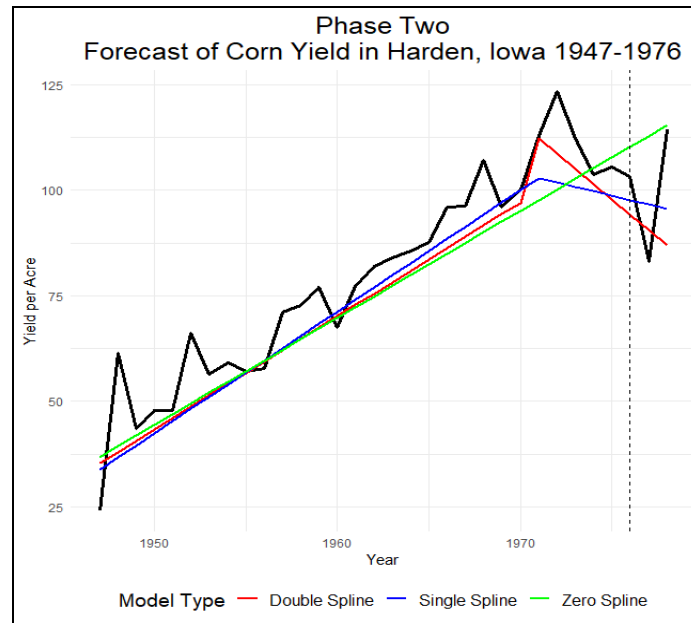


Figure 5. Visual of Regression Techniques Used in Harden, Iowa from 1947-1976.

Phase 3 - Part 1

Phase 3 of the research consisted of two parts, each designed to explore the impact of forecasting with different constraints on spline placement within the 30-year window. Both parts of Phase 3 were conducted under the RMA's method of forecasting counties individually and under the W&A's method of forecasting counties based on the risk region groupings.

In the first part of Phase 3, the aim was to extend the findings from Phase 1 and Phase 2 by expanding the constraint on the placement of breakpoints: ensuring that the final spline model does not occur within the final 10 years, and subsequently 15 years, of the 30-year window. This constraint was chosen to reduce the impacts of influential points near the end of the window. In the first two phases, findings showed that large swings in the positive or negative direction in the actual yield trend had proportionally large impacts on the forecast model. Because they occurred toward the end of the window, these deviations had large effects on the final position of the forecast line, often negatively impacting the accuracy of each model. The subsequent graph depicts the forecasts on Harden, Iowa using a 5, 10, and 15-year final spline constraint. The green, yellow, and purple vertical lines signify the 5, 10, and 15-year constraint visually while the black line separates in-sample data from out-of-sample data.

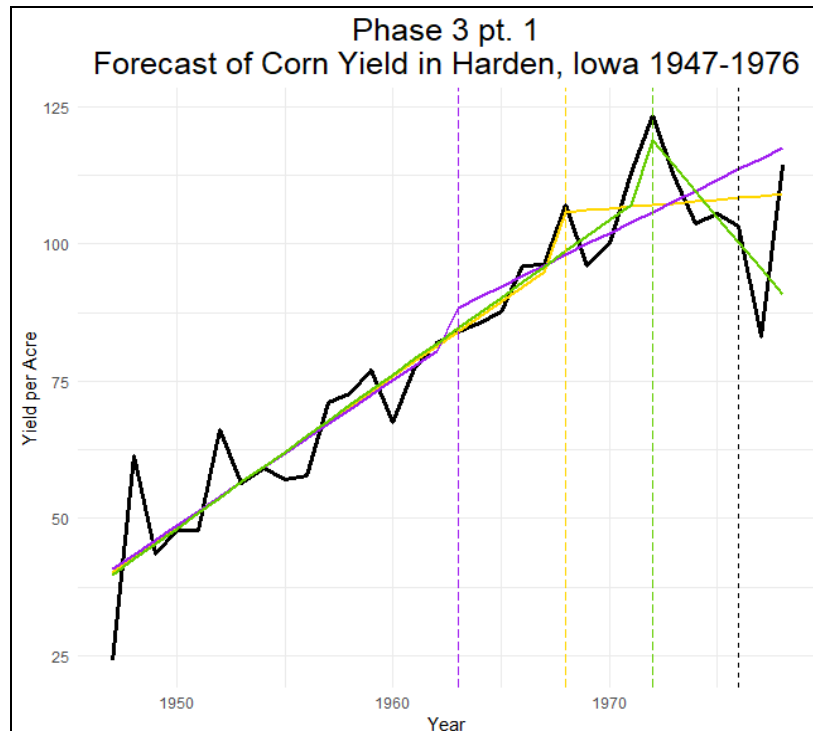


Figure 6. Visual of Forecasts Using Varying Spline Constraints.

To conduct the first part of Phase 3, the methodology employed in the previous phases was replicated, ensuring consistency and comparability of results. This involved the use of Ordinary Least Squares (OLS) regression, piecewise linear single-spline forecast, and double-spline forecast techniques as described above. The team wanted to analyze the effect of this breakpoint constraint expansion while holding as many other factors equal as possible. The results of a model that uses significantly different underlying methodology would have been meaningless, even if they exceeded the original method's results.

Once the methodology was replicated, the models were optimized using in-sample SSE as the criterion. Additionally, Akaike Information Criterion (AIC) values were computed to evaluate the aforementioned balance between accuracy and complexity. Subsequently, in the same manner as in the previous two phases, yields were forecasted for the two years following the forecast window using the optimized models. The accuracy of these forecasts was then assessed through out-of-sample backtesting.

Phase 3 - Part 2

The objective of the second part of Phase 3 was to build on the earlier findings by adding a new constraint: the slope of the regression line must be non-negative. This requirement stems from the economic assumption that agricultural technology, infrastructure, and methods are unlikely to deteriorate enough such that long-term yield trends would become negative.

To introduce this constraint, the model could not be built using Ordinary Least Squares regression, so the model was constructed using linear programming approaches in R. This methodology allows for the direct introduction of constraints into the model. Mathematically, this appeared as requiring the sum of

slopes to be greater than zero. Moving away from traditional regression techniques precluded the team from using traditional goodness-of-fit statistics, such as AIC. The residuals produced by the model are non-normal, caused by the constraint resulting in larger negative residuals, so the maximum likelihood estimator, which is normally employed to calculate the AIC, shouldn't be used. Due to the reduced effectiveness of comparing models, as well as the significant computing time necessitated by their complexity, our team chose to only look at double-spline models. Initial analysis indicated that they performed better than both single and zero spline models (both in- and out-of-sample). Finally, the forecast errors were split into corn and soybean, and the RMSFE was calculated for each group. The subsequent graph depicts the linear programming technique when used on the corn yield in Harden, Iowa from 1947-1976. The two lines indicate whether the forecast was based on the individual county or the corresponding aggregated risk region. The black dotted line indicates the separation between in-sample and out-of-sample data.

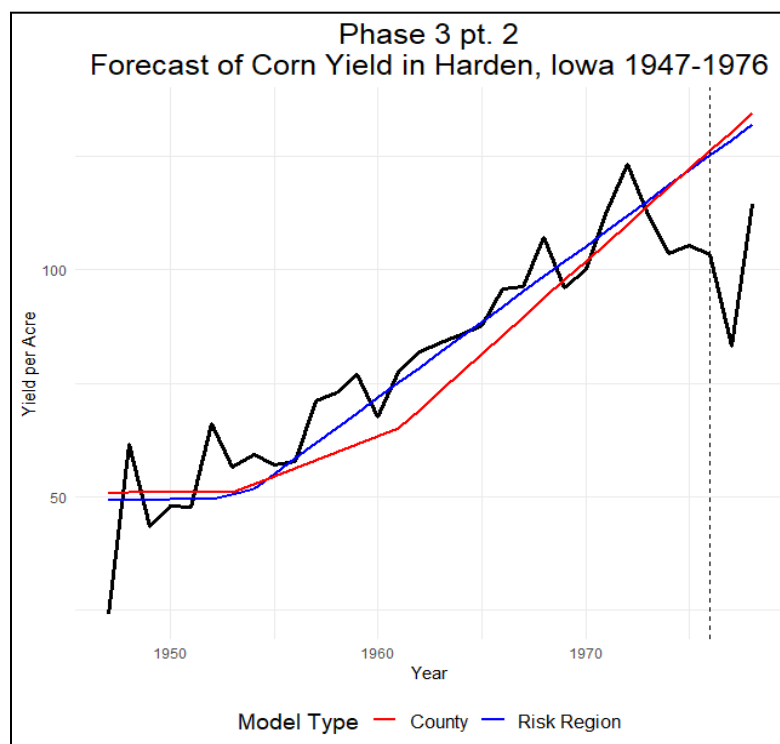


Figure 7. Visual of Linear Programming Technique Used on Harden, Iowa.

3. Data Collection

The data utilized for this project consisted of pilot data provided by Watts and Associates. The dataset had undergone cleaning and processing by their firm before being forwarded to our group. It's important to note that apart from this dataset, no additional data collection techniques were employed in the study. The analysis and experimentation relied solely on the data provided by Watts and Associates, ensuring consistency and comparability with the methodologies used by the organization. This approach allowed the focus to remain on refining existing forecasting methodologies rather than on gathering new data.

4. Data Analysis

Since the dataset used in this study was provided to us by Watts and Associates, there was no need for data validation as it had already undergone cleaning and processing by the organization. Our primary task was to familiarize ourselves with the variable definitions and the context in which the data was collected.

To begin the data analysis process, we thoroughly reviewed the documentation provided by Watts and Associates, which included descriptions of the variables. When questions arose about several aspects of the data, such as clarification of variable names or reasoning behind the inclusion of some of the data, representatives from Watts and Associates were quick to help. Their ready assistance allowed us to gain a comprehensive understanding of the dataset and its underlying structure.

Next, we conducted exploratory data analysis (EDA) to gain insights into the relationships, and patterns present within the data. This involved visualizing the data through basic plots and identifying any potential outliers or anomalies that may impact the analysis. For example, the start date of the data associated with corn yields was always 1947 while the start date for soybean data was variable.

Throughout the data analysis process, we maintained transparency and rigor in our approach, ensuring that all decisions and interpretations were based on sound statistical principles and empirical evidence. Additionally, we documented our methodology and findings to facilitate reproducibility and transparency in the research process.

5. Development of Proposed Improvement

To effectively improve upon the methods used by the RMA and Watts & Associates, we looked back to past classroom experiences and even consulted some of those who gave us such experiences. Initially, we delved into our past academic experiences from classes such as Regression, Financial Econometrics, and Economic Financial Management to consider several novel methodologies that would allow us to improve upon the models used in the first two phases. First, we considered adding more variables to the regression that are correlated to crop yields. We reasoned that by making the data more robust, as opposed to the model itself, we could find meaningful relationships between the additional agricultural factors. However, due to the delayed nature of agricultural data reporting, we were not able to find readily available data.

Next, we reviewed material from Managerial Forecasting and Decision Analysis, a course focused on the decision rules of selecting appropriate forecasting methods and the interpretation of their results. Our group considered implementing an Autoregressive Integrated Moving Average (ARIMA) model, which was unfit due to our data's complex underlying seasonality and relative irregularity. We also considered using a Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, which is typically applied to data sampled at daily or lower frequencies such as long-term financial data, to analyze underlying volatility in crop yields. However, because GARCH models require a relatively high sample size, we encountered overfitting issues which resulted in poor out-of-sample results.

After our unsuccessful attempts at choosing a sufficiently effective "classroom model," we asked our professor of Financial Econometrics, Professor Seunghyun Lee, for recommendations. Professor Lee introduced to us the concept of Major Land Resource Areas (MLRAs), which are geographical regions delineated by the USDA to characterize the landscape and its suitability for various land uses, particularly

agriculture. MLRAs are defined based on factors such as climate, geology, topography, and soil characteristics. The concept of MLRAs helps in understanding the diversity of land resources across the United States and aids in making informed decisions regarding land management, conservation, and agricultural practices. They provide a framework for organizing and interpreting information about soils, climate, vegetation, and land use potential within specific geographic regions.

MLRAs are very similar to the concept of Risk Regions developed by Watts & Associates, differing in that they have more widespread application and therefore take into account many more factors (climate, topography, land use, hydrology, biotic factors, etc.) than their agriculture-specific counterparts. Their addition to our project would be beneficial as an alternative to Risk Regions, but we deemed their implementation outside the scope of this project. Perhaps if the alternative regional delineation of MLRAs had been given earlier prioritization, their inclusion in our model could have offered beneficial improvements.

Professor Lee also exposed us to the idea of using a fixed effects ordinary least squares (FEOLS) regression model to account for correlation in county crop yields to create our improved model. However, this method necessitates more variables than we could find in any readily available government agricultural dataset.

Finally, we sought out recommendations from our esteemed and long-time professor of Engineering and Economic Financial Management, Dr. Joseph Atwood. In one of several meetings, Dr. Atwood explained how unlikely it was that evaluating yields on a 5-year basis would result in the best forecast. He emphasized the importance of considering longer-term trends and variability in agricultural systems, suggesting that a more comprehensive analysis spanning larger time windows could provide a more robust understanding and accurate prediction of yields. Dr. Atwood advocated for extending the window where the ending spline could not occur (within the first or last five years), which would minimize the overfitting of the end of the data by the regression model. Dr. Atwood also suggested incorporating a nonnegativity constraint to the slopes of the final splines in the regression. He reasoned that this constraint would leverage the assumption that crop yields are unlikely to decrease in the future, but rather they will either stabilize or continue their upward trend. This recommendation aligns with the notion that advancements in technology and improvements in agricultural infrastructure tend to support ongoing productivity gains in crop production, holding other factors constant.

6. Proposed Solution 1

The team opted to implement the results of the consultation with Dr. Atwood, both of which require further elaboration. The first pivotal improvement the team introduced involved strategically adjusting the placement constraints of splines within the regression models, with a particular focus on the final spline. Traditionally, the final spline in current models is allowed within a relatively narrow window of the end, typically extending beyond the last five years of the dataset. However, recognizing the need for a more nuanced approach to capturing agricultural dynamics, the team implemented a reduction in the permissible range for spline placement. Specifically, the constraint was modified so that the final spline must be positioned further from the end of the dataset, with an added minimum distance of 10 to 15 years.

This adjustment was driven by multiple rationales rooted in the unique characteristics of agricultural systems and the intricacies of crop yield data, specifically its long time horizon. By requiring the final spline to be positioned further from the end of the dataset, we aimed to mitigate the potential for overfitting and enhance the generalizability of the regression models. Overfitting, a common challenge in

regression analysis, occurs when a model captures noise or random fluctuations in the data rather than the underlying relationships. Furthermore, by imposing stricter constraints on spline placement, it encourages the identification of more stable and enduring patterns in crop yield dynamics. By anchoring the final spline to data points further removed from the present, our models prioritize the capture of long-term trends and mitigate the influence of short-term fluctuations or noise.

7. Proposed Solution 2

As discussed above, a second notable improvement the team introduced lies in the fundamental structure of the regression models. Recognizing the inherent nature of agricultural production, where technological advancements, agronomic practices, and environmental factors predominantly contribute to crop yields, the team added the modification that all final slopes within the regression framework must be non-negative.

This adjustment stems from a nuanced understanding of the agricultural landscape, where the trajectory of crop yields is characterized by a persistent upward trend. Historically, advancements in agricultural science and technology have consistently propelled yields upwards, offsetting the occasional setbacks posed by environmental factors or other challenges. Therefore, constraining the slopes to be non-negative aligns with the empirical reality that agricultural productivity seldom exhibits a declining trend. By mandating non-negative slopes, our regression models encapsulate the intrinsic growth pattern of crop yields. Moreover, this modification fosters interpretability, as it aligns with the intuitive understanding that technological innovations and agricultural interventions typically lead to yield enhancements.

8. Results

Phase 1 - Results

In Phase 1 of the study, the Risk Management Agency methodology was employed to forecast crop yields, focusing on corn and soybean. The Root Mean Squared Forecast Error (RMSFE) statistics provided insights into the accuracy of the forecasting approach.

For corn, the RMSFE exhibited a wide range of values, with the minimum observed RMSFE at 0.01 bushels per acre, indicating highly precise forecasting in certain instances. However, the maximum RMSFE reached 189.97 bushels per acre, indicating significant forecasting errors in specific scenarios. On average, the RMSFE for corn stood at 29.48 bushels per acre (20.9% error) suggesting a moderate level of forecasting accuracy across the dataset.

For soybeans, similar patterns were observed. The minimum observed RMSFE was 0.05 bushels per acre, indicating instances of highly accurate forecasting. However, the maximum RMSFE reached 53.19 bushels per acre, indicating significant forecasting difficulty in certain cases. The average RMSFE for soybean was 12.67 bushels per acre (56.24% error), suggesting a slightly lower level of forecasting precision on average compared to corn.

Visuals and summary statistics are provided in the below graphs and table.

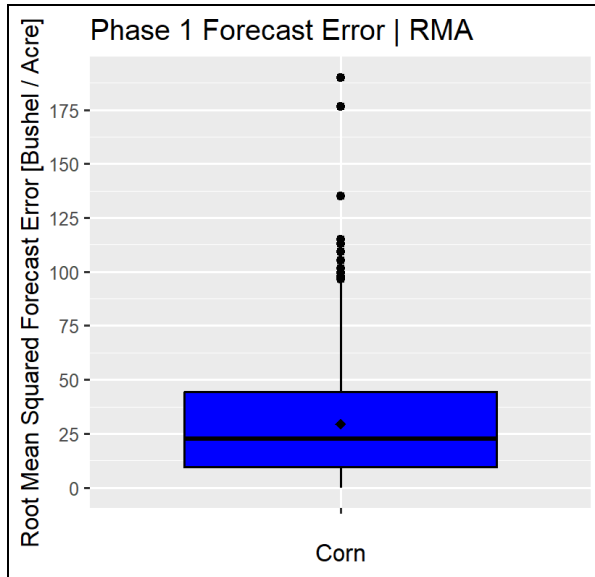


Figure 8. RMSFE on Corn in Phase 1.

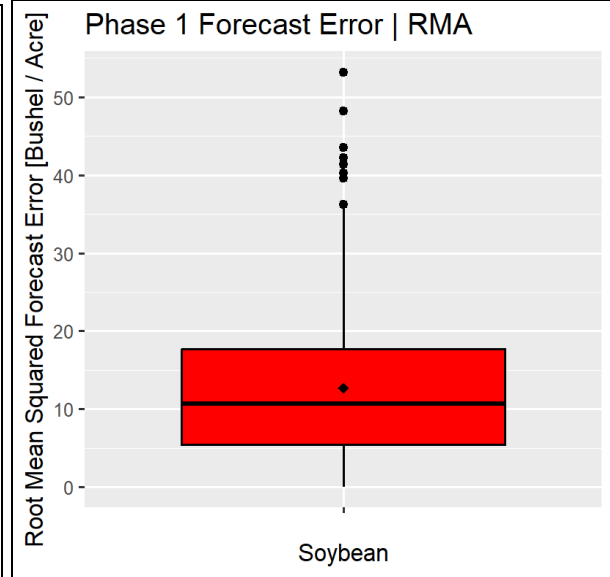


Figure 9. RMSFE on Soybean in Phase 1.

Table 1. Summary Statistics on the RMSFE for Corn and Soybean in Phase 1.

Crop	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
Corn	0.01363636	9.761574	29.47860	22.69598	44.29290	189.97
Soybean	0.04615370	5.479155	12.67159	10.72460	17.73375	53.19

Phase 2 - Results

In Phase 2, the objective was to enhance comprehension of and replicate the methodologies used by Watts & Associates for forecasting crop yields, with a focus on corn and soybean. The Root Mean Squared Forecast Error (RMSFE) statistics were analyzed to evaluate the accuracy of the forecasting approach.

For corn forecasting, the RMSFE statistics revealed a range of accuracy levels. The minimum observed RMSFE was 0.05 bushels per acre, indicating highly precise forecasting in some instances. However, the maximum RMSFE reached 107.62 bushels per acre, highlighting significant forecasting errors in specific scenarios. On average, the RMSFE for corn stood at 22.70 bushels per acre (16.4% error), suggesting a moderate level of forecasting accuracy across the dataset.

For soybean forecasting, a similar pattern emerged. The minimum observed RMSFE was 0.03 bushels per acre, indicating instances of highly accurate forecasting. However, the maximum RMSFE reached 48.29 bushels per acre, indicating significant forecasting errors in certain cases. The average RMSFE for soybeans was 12.10 bushels per acre (55.75% error). The percent error was much larger than that of corn again for Phase 2, suggesting a lower level of forecasting precision on average compared to corn.

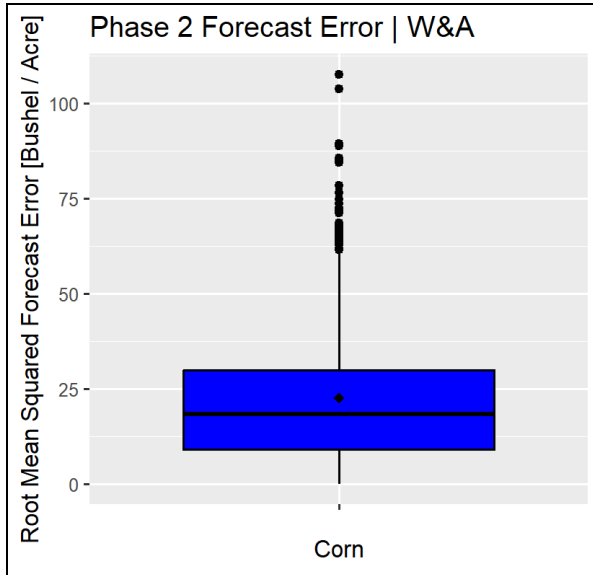


Figure 10. RMSFE on Corn in Phase 2.

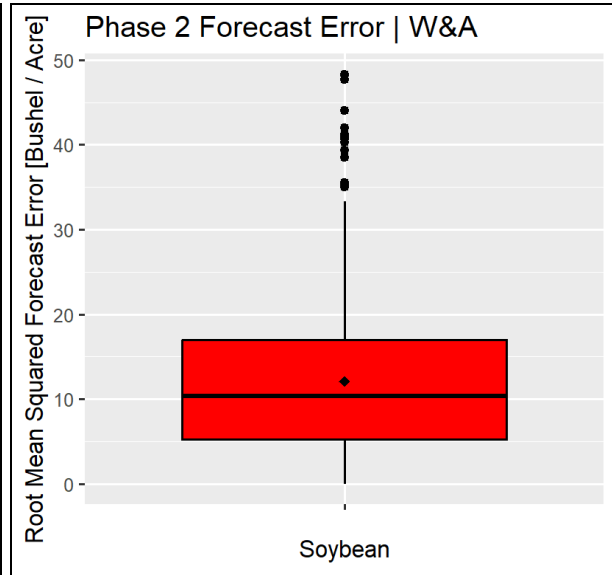


Figure 11. RMSFE on Soybean in Phase 2.

Table 2. Summary Statistics on the RMSFE for Corn and Soybean in Phase 2.

Crop	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
Corn	0.04806572	9.007275	22.70210	18.38528	29.98888	107.6224
Soybean	0.02603539	5.294801	12.10148	10.33296	16.97153	48.2922

Phase 3 Part 1 - Results

Phase 3 of the study advanced the exploration of forecasting methodologies, focusing on refining accuracy through the introduction of additional constraints. This phase aimed to optimize forecasting models by varying spline restrictions and implementing constraints such as non-negativity of the slope of the regression line.

The research divided Phase 3 into two parts. In the first part, the study investigated the impact of different spline restrictions on forecasting accuracy for both individual counties and Risk Regions. The results unveiled nuanced variations in forecasting performance, highlighting the importance of spline placement within the 30-year window. This analysis accounted for differences between Risk Regions and individual counties, providing insights into the effectiveness of different methodologies across distinct geographic scales. The results are presented in the tables and graphs in Appendix B through Appendix E.

Appendix B demonstrates the results of the RMA's methodology applied to corn yields. On average, we observe decreased forecast errors as the impermissible spline window is expanded from 5 years to 10 years and subsequently to 15 years. The smallest window of 5 years produces a mean error of 29.48 bushels per acre (20.91% error), which falls to 21.08 (14.95% error) and 18.60 bushels per acre (13.16%

error) with 10- and 15-year windows, respectively. While minimum error values are negligibly equal, we also see a decreased maximum error as we increase the spline window.

In Appendix C, the results of the RMA's individual county method applied to soybeans are provided, which closely follow those in Appendix A. On an individual county basis, mean soybean yield forecast error fall from 12.67 (56.25% error) to 9.44 (45.43% error) to 8.04 bushels per acre (38.62% error) with 5-, 10-, and 15-year windows, respectively. Additionally, we again observe smaller maximum errors with increased windows.

Appendix D shows error rates for corn when analyzing W&A's Risk Regions under the lens of lengthening the final segment. Similar to those of individual county corn yields, these average means fall the further back a spline is allowed to be placed, from 22.70 (16.4% error) to 19.64 (13.77% error) and again to a slightly smaller error of 19.18 bushels per acre (13.50% error).

Lastly, in Appendix E, which shows aggregated Risk Region forecast error on soybean yields after expanding the final spline's range of restriction, show similar results to the first three analyses. Under a 5-year window, we observe an error of 12.10 bushels per acre (55.75% error). Consistent with other crop/method combinatorial results, Appendix D shows a 10-year error of 9.20 bushels per acre (45.26% error) and a 15-year error of 8.22 bushels per acre (40.62%).

The results of Phase 3 Part 1 are consistent and clear for both corn and soybeans as well as for both the RMA's individual county method and for Watts and Associates' Risk Region method. When the size of the allowable range for the final spline to be located is expanded from a 5-year window to a 10- and 15-year window, forecast errors decrease and prediction appears to become more accurate. Though more thorough knowledge of the data and relationships between its factors is necessary for a deeper analysis, our team postulates that a spline located further from the end of the forecast window has more in-sample data to analyze. As such, it would be able to capture more variation in the later observations of actual yield, explaining the seemingly positive relationship between a larger window and forecast accuracy.

Phase 3 Part 2 - Results

Building upon the findings of the first part, Phase 3 Part 2 explored the effectiveness of additional constraints, such as ensuring non-negativity of slope, in further refining forecasting accuracy. This involved employing linear programming techniques to optimize forecasting models by constraining the slope after the final spline into always being non-negative. Similar to Part 1, Part 2 considered the differences between Risk Regions and individual counties, enabling a comprehensive evaluation of forecasting methodologies across various spatial scales. The results are presented in the tables and graphs in Appendix F through Appendix I.

Appendix F and G present the summary statistics on the forecast error when linear programming was used on the individual county to forecast yield. With a 5-year spline constraint, an 18.3 (13.39% error) bushel per acre forecast error was measured on corn yields. For soybean yields, a 7.46 (34.2% error) bushels per acre forecast error was measured.

Appendix H and I present the summary statistics on the forecast error when linear programming was used at the risk region level to forecast yield. With a 5-year spline constraint, a 19.5 (14.2% error) bushel

per acre forecast error was measured on corn yields. For soybean yields, a 7.47 (34.0% error) bushels per acre forecast error was measured.

9. Economic Analysis

While a detailed and accurate analysis of the trickle-down economic effect of improved forecasts is infeasible to arrive at and outside the scope of the team's research, a general theoretical economic framework is discussed.

Should our novel improvements on incumbent models be implemented, assuming they remain valid and viable across other county and crop combinations, we would expect to see a cascade of benefits eventually trickling down to the end consumer, our most important stakeholder. Enhanced accuracy of yield predictions could reduce the financial risks associated with crop production, meaning farmers can better anticipate crop outlooks and plan accordingly, leading to improved efficiency and lower production costs. Consequently, farmers can mitigate the need for costly risk management measures, such as excessive pesticide or fertilizer usage, contributing to overall sustainability in agricultural practices.

Additionally, associated cost savings realized by farmers extend to crop insurance providers who experience reduced payout obligations due to more accurate risk assessment and pricing. By minimizing losses associated with yield fluctuations, insurers can offer more competitive premiums to farmers. With lower production and risk mitigation costs for producers, there is less pressure to pass on increased expenses to consumers in the form of higher prices for their crops. As a result, households experience greater affordability and purchasing power at the grocery store. These cost savings stimulate economic activity and promote overall prosperity, demonstrating the far-reaching impact of accurate crop yield forecasting on the overall economy.

10. Stakeholder Presentation

Because of the far-reaching effects of effective and efficient crop insurance on various entities spanning the country and globe, the team has identified several stakeholders with vested interest in the outcome of our project.

First, and perhaps most importantly, end consumers of food worldwide represent a significant group of stakeholders in this crop insurance project. With a direct interest in food security, affordability, and quality, consumers rely on stable agricultural production to meet their dietary needs. According to the Kassel (2024) of the USDA, of the \$278.2 billion in total crop cash receipts in 2022, corn and soybean production was responsible for over 53%, or \$148.5 billion. For consumers of these crops, the effectiveness of crop insurance - more specifically, the accuracy of yield forecasts - directly impacts the stability of food prices and the accessibility of essential commodities. Our ability to offer accurate forecasts to insurance firms translates to supply stability and reduced price volatility for such insurers. The benefits of this are twofold; first, firms keeping their costs down can avoid the pressure of raising their prices. Second, the government may choose to offer less subsidy to healthy insurance firms, resulting in lower taxes for the American public, as will be explained. With a more predictable supply chain, producers can better manage their inventory and production costs, thereby exerting downward pressure on prices at the retail level. As a result, consumers benefit from lower grocery bills and increased purchasing power, translating into potential savings on household expenses.

Next, Watts & Associates, our project sponsor, is a prominent crop insurance consultancy that holds a significant stake in our research. While they do not directly provide crop insurance, W&A plays a crucial

role as advisors to crop insurance companies, offering expertise in risk assessment, policy development, and regulatory compliance. As stakeholders, W&A is invested in advancing the efficacy and sustainability of crop insurance programs, ensuring they meet the needs of both agricultural producers and insurance providers.

Thirdly, the clients of W&A, which are mainly insurance firms providing crop coverage, hold a vested interest in our work. According to a 1991 by the US General Accounting Office, yield overestimates generally result in lower premium payments to farmers. This may raise no complaints from the insurance firms, but if farmers perceive yields as too low, participation from farmers may fall off as they become discontented and disincentivized. Thus, low forecasted yields would lead to decreased client counts for insurance firms. Conversely, overestimated crop yields result in greater indemnity payments from insurers to farmers, which bodes ill for the providers of crop insurance. In this way, consumers, farmers, insurance consultants, and insurance providers all stand to benefit from increased accuracy in the forecasts of corn and soybean yields, which our team seeks to improve.

11. Project Management

The foundation of our project management strategy lies in meticulous planning, which commenced during the project's conceptualization phase. At the outset, the research objectives were clearly defined, delineating the scope, key stakeholder interests, and focus of the study. These objectives served as guiding principles, informing subsequent decisions and actions throughout the research process.

To translate these objectives into actionable steps, sprint planning was conducted in Yodiz. The project was broken down into two-week sprints where tasks, milestones, and deliverables were defined at the beginning of the sprint, and a summary of the execution of the plan as well as the outlook was reported on at the end of the sprint. Key considerations included the allocation of responsibilities among team members, the establishment of timelines for task completion, and the identification of potential risks and contingencies.

Effective team coordination was paramount to the success of the project. Assembling a multidisciplinary team with expertise in areas such as agricultural economics, statistics, and data analysis was crucial to ensure comprehensive coverage of the research objectives. Each team member was capable due to the Financial Engineering curriculum. Multiple team meetings were held weekly to facilitate communication, foster collaboration, and address any challenges or bottlenecks encountered during the research process.

Furthermore, clear lines of communication were established to facilitate seamless information exchange among team members. This included designated channels for sharing updates, progress reports, and relevant research findings. By promoting open communication and collaboration, team members were able to leverage their collective expertise and contribute synergistically to the research effort.

Adhering to predefined timelines was essential to ensure the timely completion of project milestones and deliverables. To this end, a project schedule was developed, outlining the sequence of tasks and their respective deadlines. This schedule served as a reference point for monitoring progress and identifying any deviations from the planned timeline.

Regular progress reviews were conducted to assess the status of ongoing tasks and identify any potential delays or obstacles. These progress reports were shared with the team's advisor Dr. Andrew Hill after each sprint. In instances where adjustments to the timeline were necessary, proactive measures were taken to mitigate risks and reallocate resources as needed. By maintaining a vigilant focus on timeline management, we were able to stay on track and achieve timely completion of project objectives.

Effective project management is indispensable to the success of any research endeavor, serving as the linchpin that ensures the seamless execution of research activities and the achievement of project objectives. In this section, we have outlined the project management approach adopted throughout the course of this research, highlighting key components such as project planning, team coordination, timeline management, resource allocation, and risk management. By adhering to these principles and practices, we were able to navigate the complexities of the research process effectively and achieve the desired outcomes of our study.

12. Conclusion

While current forecasting techniques work well enough to remain in existence, the design team sees room for improvement with the implementation of alternative methods to forecast agricultural yields. First, creating Risk Regions composed of counties with similar historical crop output appeared to decrease average error rates. In terms of corn, average errors of 20.9% under the RMA's incumbent model dropped to 16.4% under Risk Regions. Similarly, for soybeans, average errors of 56.24% under individual counties fell slightly to 55.75%. These results suggest that if the implementation of these models across the plethora of other crops, county, and year combinations proves viable, the RMA's longstanding approach of individual counties may be superseded by the Risk Region technique developed by Watts & Associates.

Next, the design team sees promise in replacing the current technique with an alternative in terms of the location of the final spline. Presently, the final spline of any forecast is restricted to being located at least 5 years before the end of the forecast window. By increasing this restriction to 10 and 15 years, the design team consistently saw decreased error rates in both corn and soybean yields. When the final spline is only restricted to not occurring in the final 5 years, the model fits the in-sample data well, but the underlying trend of the data often is neglected - leading to inaccuracies. However, when the final spline is moved back to 10 or 15 years, the underlying trend is captured well and the accurate forecasts support this. As above, should this result hold across other national combinations, the team believes that this spline window alternative could disrupt the current system.

Finally, when accommodating the assumption that long-term yield trends cannot be negative, we again observed decreased error rates on average. By introducing the non-negativity constraint on the trend line of individual county data, a 13.39% and 34.2% error rate was observed in corn and soybean yields, respectively, on a 5-year horizon. Compared to the initial 20.9% and 56.24% rates seen under the current model, this represents a significant potential improvement. Not surprisingly, the same trend occurs under Risk Regions at the 5-year level. The resulting 14.2% error for corn yield and 34.0% error for soybean yield are quite promising when compared with the corresponding 16.4% and 55.75% errors of soybean yields grouped by Risk Regions.

The group's attempt at making minor changes to the current models used by the Risk Management Agency and Watts and Associates aimed to streamline implementation. The group did not have access to the code used by the RMA or W&A, but adjusting the final spline constraint should be a minor adjustment that requires minimal additional effort. While the linear programming technique used in Part

2 of Phase 3 may necessitate a more substantial investment to implement, the team chose to continue using the current models' approaches with individual county and risk region forecasts.

The results of both the forecasting methods used by the group suggest that there is clear room for improvement by the Risk Management Agency and, if the forecasting methodology applies to all crop-county combinations across the nation, the two novel methods considered in this project could reduce the errors. The project was confined to the narrow scope of corn and soybean yields in three states, so testing the methods on a wider demographic of states and crops would be valuable.

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Appendices

Appendix A | Academic Course References and Utilization

Table 3. Table of Course Content Utilized for Capstone Project.

Course Number	Course Name	Utilization
EFIN 401	Engineering & Economic Financial Management II	- Out-of-sample testing - Linear Programming - Analysis of Results
ECNS 461	Financial Econometrics	- Regression Techniques - Time Series Analysis
EIND 468	Managerial Forecasting & Decision Analysis	- Regression Techniques - Time Series Analysis - Measuring Forecast Errors - Visualization in R Studio

Appendix B | Phase 3 Part 1 Measured Forecast Errors on Corn by Individual County

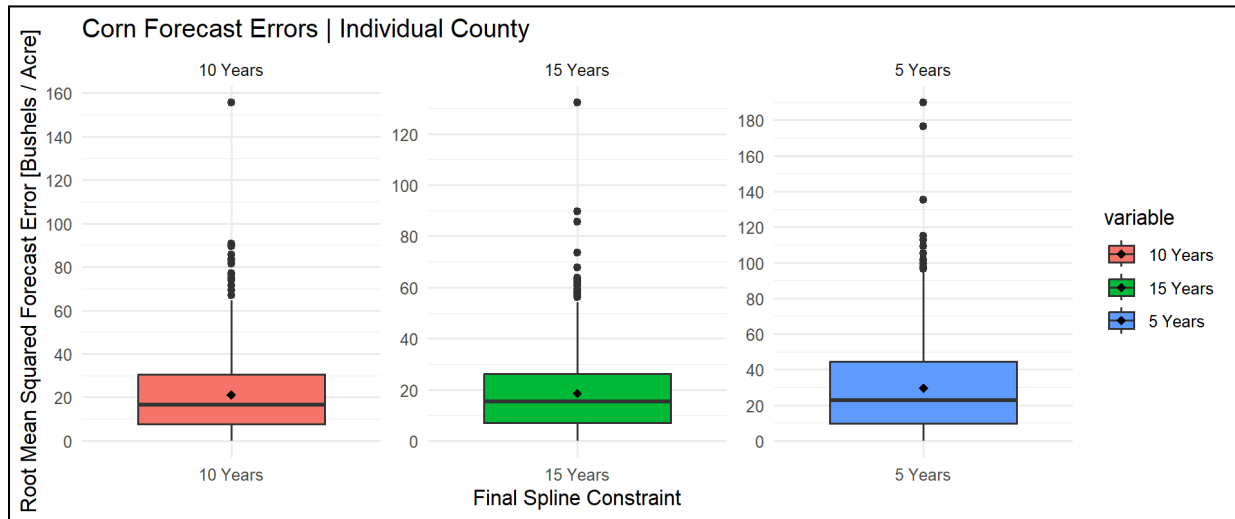


Figure 12. Boxplot of Corn RMSFE using Individual County Forecasts.

Table 4. Summary Statistics of Corn RMSFE using Individual County Forecasts [bushels/acre].

Spline_Constraint_Year	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
5 Year	0.01363636	9.761574	29.47860	22.69598	44.29290	189.9700
10 Year	0.01363636	7.538393	21.08476	16.43167	30.48720	155.6889
15 Year	0.02782403	7.105208	18.60346	15.40555	26.12577	132.3042

Appendix C | Phase 3 Part 1 Measured Forecast Errors on Soybeans by Individual County

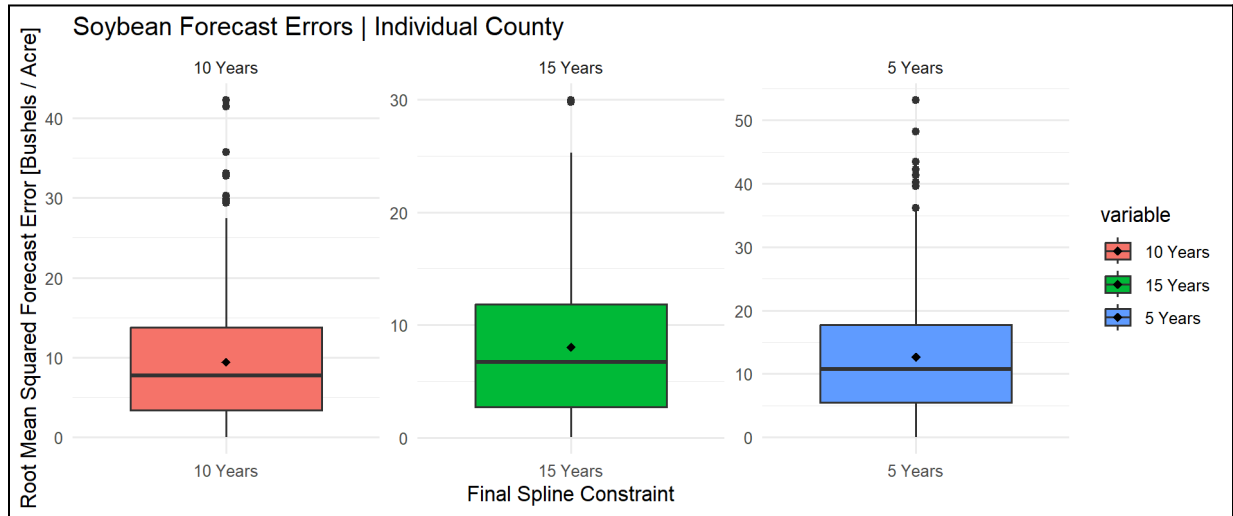


Figure 13. Boxplot of Soybean RMSFE using Individual County Forecasts.

Table 5. Summary Statistics of Soybean RMSFE using Individual County Forecasts [bushels/acre].

Spline_Constraint_Year	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
5 Year	0.0461537	5.479155	12.671587	10.724601	17.73375	53.19000
10 Year	0.0461537	3.410513	9.439620	7.710075	13.76773	42.25454
15 Year	0.0461537	2.711218	8.044267	6.688072	11.83021	29.96937

Appendix D | Phase 3 Part 1 Measured Forecast Errors on Corn by Risk Regions

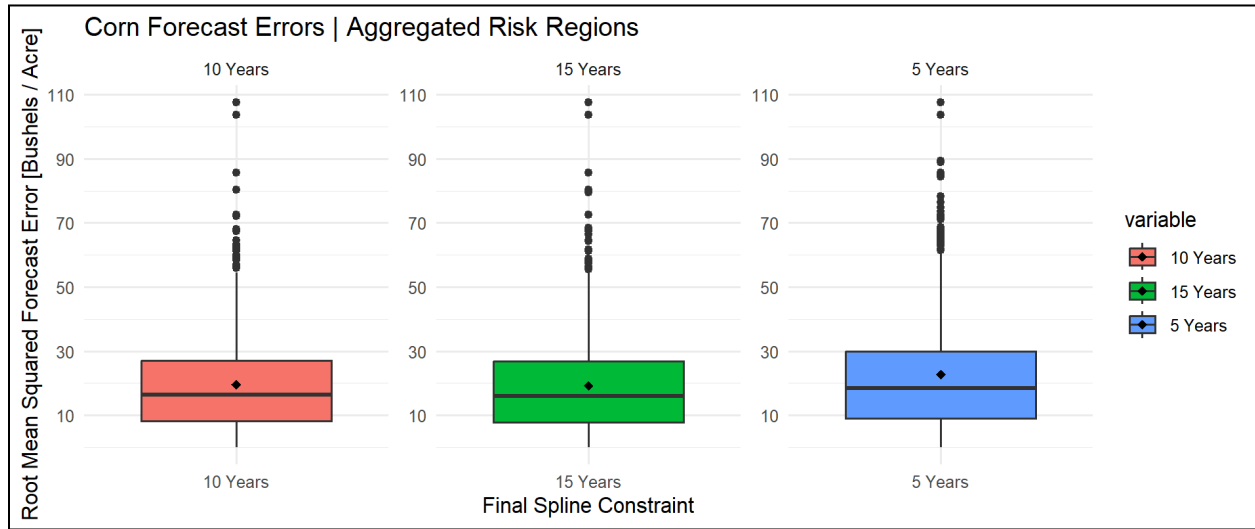


Figure 14. Boxplot of Corn RMSFE using Aggregated Risk Region Forecasts.

Table 6. Summary Statistics of Corn RMSFE using Aggregated Risk Region Forecasts [bushels/acre].

Spline_Constraint_Year	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
5 Year	0.04806572	9.007275	22.70210	18.38528	29.98888	107.6224
10 Year	0.03025924	8.220654	19.63855	16.47879	27.05436	107.6224
15 Year	0.04806572	7.777029	19.18163	16.00596	26.89043	107.6224

Appendix E | Phase 3 Part 1 Measured Forecast Errors on Soybeans by Risk Regions

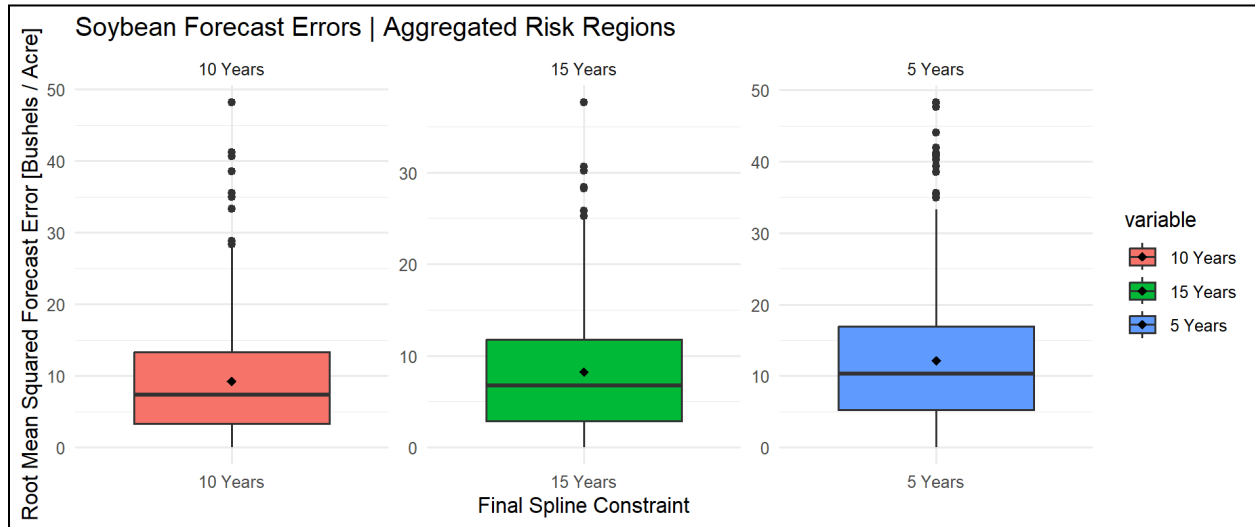


Figure 15. Boxplot of Soybean RMSFE using Aggregated Risk Region Forecasts.

Table 7. Summary Statistics of Soybean RMSFE using Aggregated Risk Region Forecasts [bushels/acre].

Spline_Constraint_Year	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
5 Year	0.02603539	5.294801	12.101484	10.332957	16.97153	48.29220
10 Year	0.02603539	3.345445	9.196549	7.316017	13.32171	48.19368
15 Year	0.02603539	2.860475	8.219261	6.739502	11.78571	37.64090

Appendix F | Phase 3 Part 2 Forecast Error with Linear Programming on Corn by Individual County

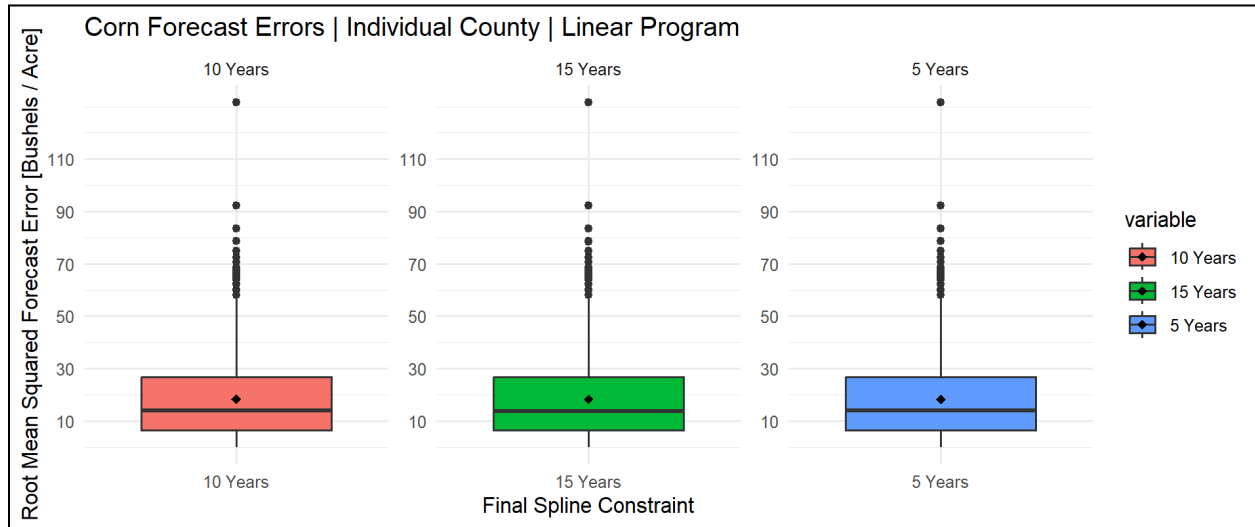


Figure 16. Boxplot of Corn RMSFE using Linear Programming on Individual County Forecasts.

Table 8. Summary Statistics of Corn RMSFE using Linear Programming on Individual County Forecasts [bushels/acre].

Spline_Constraint_Year	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
5 Year	0.001716456	6.440031	18.28457	13.87686	26.83813	131.6561
10 Year	0.001716456	6.440031	18.28457	13.87686	26.83813	131.6561
15 Year	0.005857648	6.463579	18.32873	13.86562	26.83813	131.6561

Appendix G | Phase 3 Part 2 Forecast Error with Linear Programming on Soybeans by Individual County

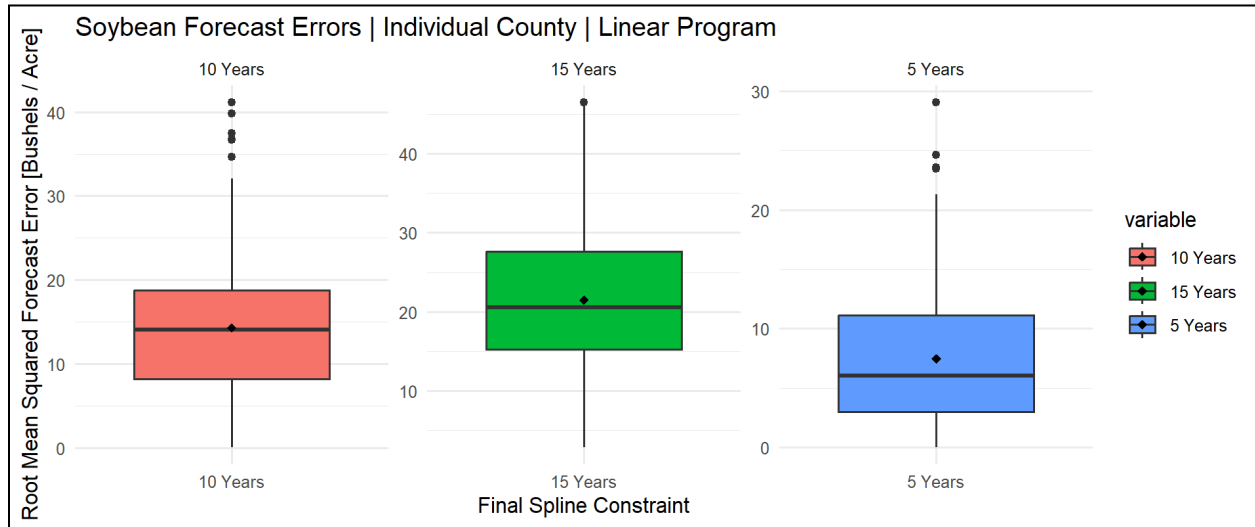


Figure 17. Boxplot of Soybean RMSFE using Linear Programming on Individual County Forecasts.

Table 9. Summary Statistics of Soybean RMSFE using Linear Programming on Individual County Forecasts [bushels/acre].

Spline_Constraint_Year	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
5 Year	0.02094540	3.020727	7.462635	6.030373	11.12757	29.04984
10 Year	0.09192497	8.227584	14.316400	14.080875	18.78050	41.15210
15 Year	2.88495875	15.219448	21.513014	20.566203	27.61351	46.47920

Appendix H | Phase 3 Part 2 Forecast Error with Linear Programming on Corn by Risk Region

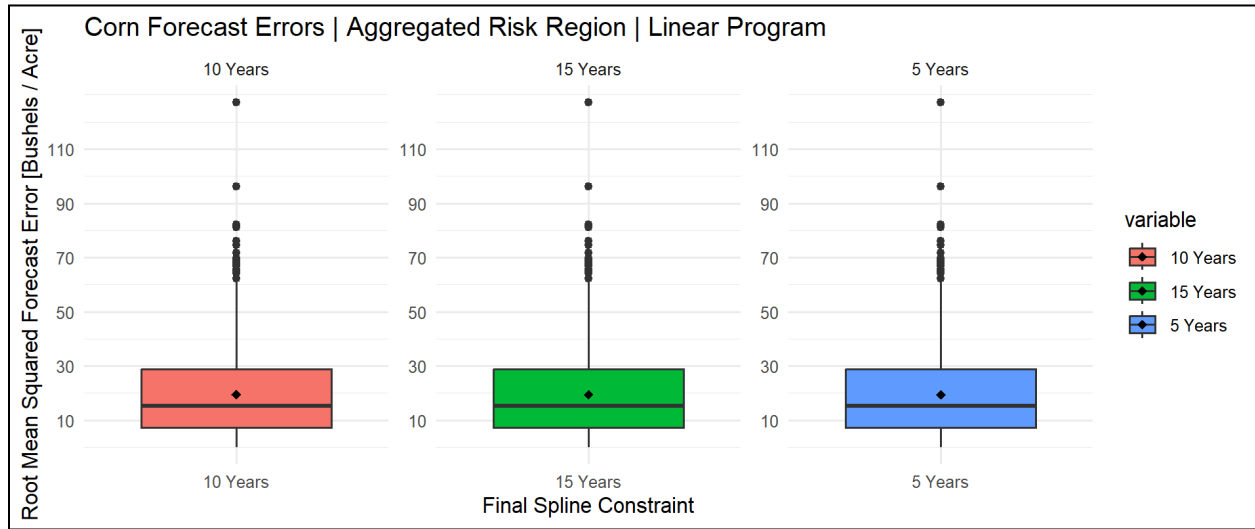


Figure 18. Boxplot of Corn RMSFE using Linear Programming on Risk Region Forecasts.

Table 10. Summary Statistics of Corn RMSFE using Linear Programming on Risk Region Forecasts [bushels/acre].

Spline_Constraint_Year	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
5 Year	0.03155075	7.170446	19.46979	15.15348	28.73388	127.2193
10 Year	0.03155075	7.170446	19.46979	15.15348	28.73388	127.2193
15 Year	0.03155075	7.170446	19.48236	15.18366	28.73388	127.2193

Appendix I | Phase 3 Part 2 Forecast Error with Linear Programming on Soybeans by Risk Regions
[Bushels / Acre]

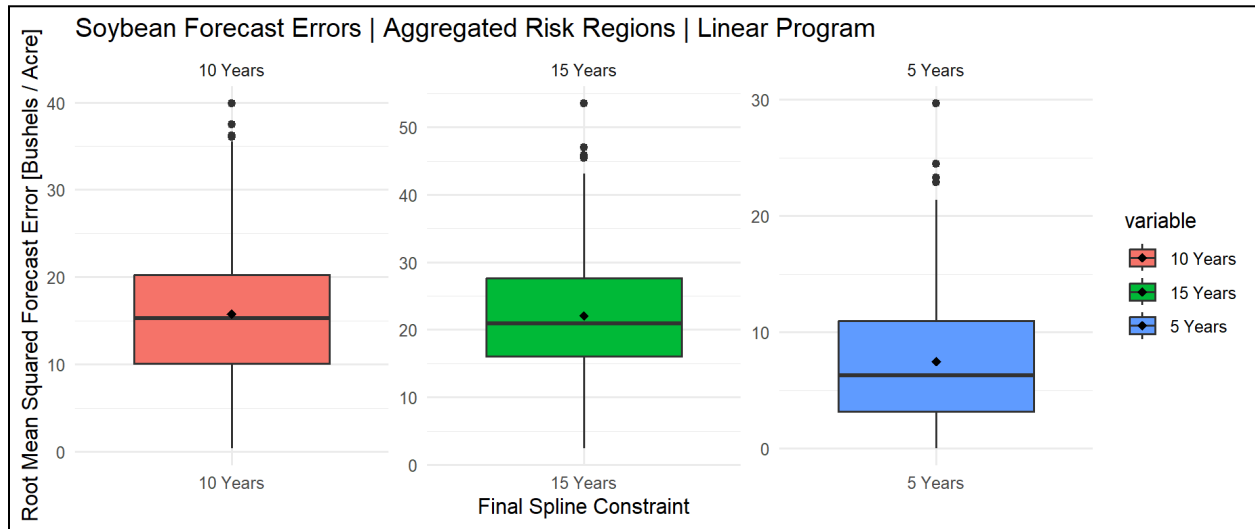


Figure 19. Boxplot of Soybean RMSFE using Linear Programming on Risk Region Forecasts.

Table 11. Summary Statistics of Soybean RMSFE using Linear Programming on Risk Region Forecasts [bushels/acre].

Spline_Constraint_Year	Min_value	Quantile_1	Mean	Median	Quantile_3	Max
5 Year	0.02446895	3.149319	7.476554	6.301093	10.98554	29.67646
10 Year	0.40122114	10.081051	15.755908	15.266784	20.27415	39.89834
15 Year	2.44905863	16.048330	22.058858	20.879709	27.61699	53.53135